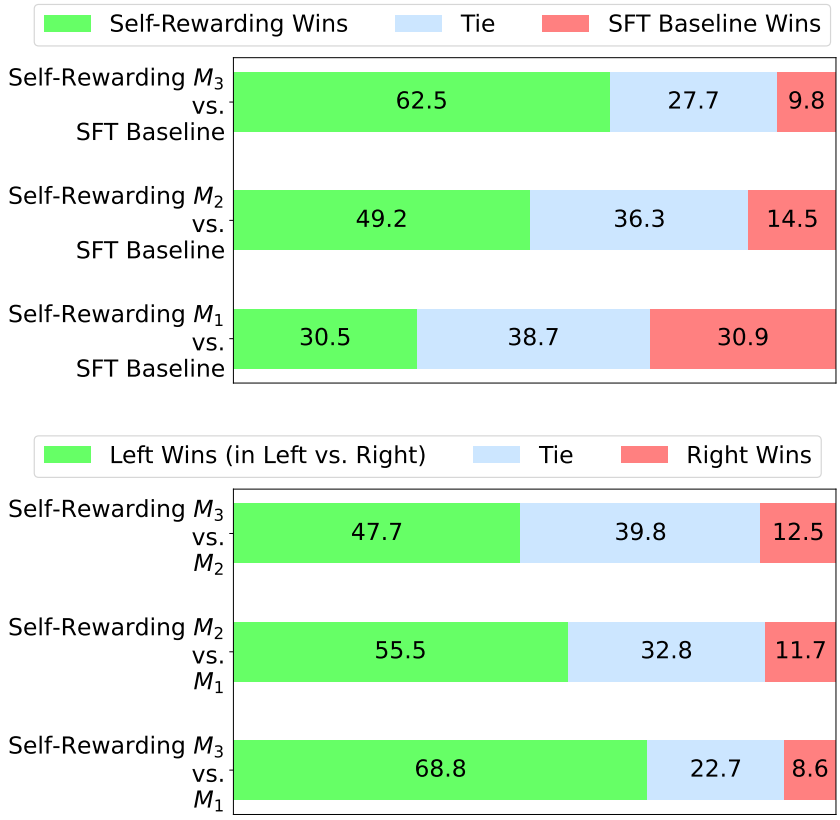

Figure 3: **Instruction following ability improves with Self-Training:** We evaluate our models using head-to-head win rates on diverse prompts using GPT-4. The SFT Baseline is on par with Self-Rewarding Iteration 1 (M_1). However, Iteration 2 (M_2) outperforms both Iteration 1 (M_1) and the SFT Baseline. Iteration 3 (M_3) gives further gains over Iteration 2 (M_2), outperforming M_1, M_2 and the SFT Baseline by a large margin.

selected models from the leaderboard in the table. We note that many of those competing models contain either proprietary alignment data (which is typically large, e.g., over 1M annotations in Touvron et al. [2023]) or use targets that are distilled from stronger models. In contrast, our Self-Rewarding model starts from a small set of seed data from Open Assistant, and then generates targets and rewards from the model itself for further iterations of training.

Fine-grained analysis As described earlier, the overall performance of the model in AlpacaEval improves with each iteration of training. It would be interesting to break down the overall performance improvement to see exactly what type of tasks these improvements come from. Therefore, we cluster the instructions in AlpacaEval test set into different groups based on three perspectives: (1) instruction category (2) instruction complexity (3) expected response length. We achieve this by using GPT-4. The detailed statistical information of the breakdown and the prompting techniques we used for getting this breakdown can be found in Appendix A.6. Results for the instruction category are given in Figure 4, and the other two in Appendix Figure 11. From the results we can conclude that (i) Self-Rewarding models can substantially improve the win rate in most categories, but there are some tasks for which this approach does not improve, such as mathematics and logical reasoning, indicating that our current training approach mainly allows the models to better utilize their existing knowledge. (ii) Through Self-Rewarding model training, the model's win rate increases on almost all tasks of different complexity, and especially on slightly more difficult tasks (complexity of 5, 6, 7 out of 10). (iii) The models also show a steady increase in the win rate on tasks with instructions with different expected response lengths.

Table 1: **AlpacaEval 2.0 results** (win rate over GPT-4 Turbo evaluated by GPT-4). Self-Rewarding iterations yield improving win rates. Iteration 3 (M_3) outperforms many existing models that use proprietary training data or targets distilled from stronger models.

		Alignment Targets	
Model	**Win Rate**	Distilled	Proprietary
Self-Rewarding 70B			
Iteration 1 (M_1)	9.94%		
Iteration 2 (M_2)	15.38%		
Iteration 3 (M_3)	20.44%		
Selected models from the leaderboard			
GPT-4 0314	22.07%		✓
Mistral Medium	21.86%		✓
Claude 2	17.19%		✓
Gemini Pro	16.85%		✓
GPT-4 0613	15.76%		✓
GPT 3.5 Turbo 0613	14.13%		✓
LLaMA2 Chat 70B	13.87%		✓
Vicuna 33B v1.3	12.71%	✓	
Humpback LLaMa2 70B	10.12%		
Guanaco 65B	6.86%		
Davinci001	2.76%		✓
Alpaca 7B	2.59%	✓	

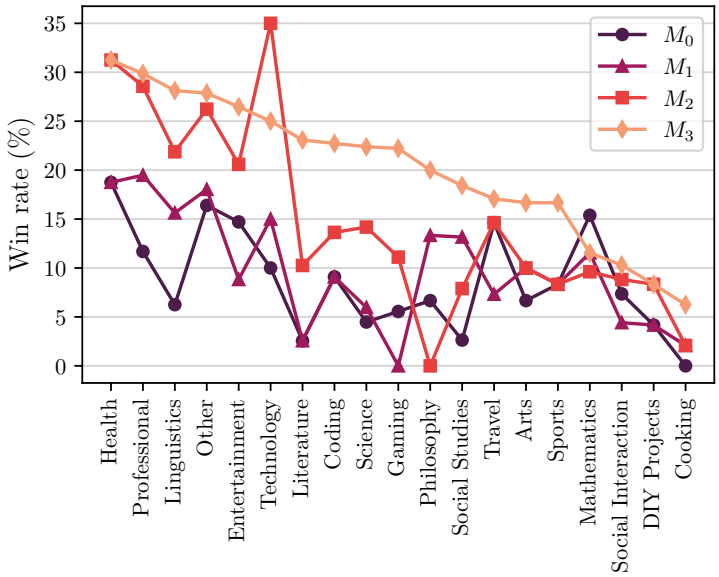

Figure 4: AlpacaEval win rate breakdown for instruction categories (full names given in Appendix). Self-Rewarding models give gains across several topics, but tend to e.g. give less gains on mathematics and reasoning tasks.

Data distribution analysis We perform a t-SNE [Van der Maaten and Hinton, 2008] visualization of the IFT, EFT and AIFT(M_1) data, shown in Appendix A.1. We find good overlap between the IFT and AIFT(M_1) examples, which is desired, while the EFT examples lie in a different part of the embedding space, which can help explain why they would not affect IFT performance. We observe that generations from M_1 on AlpacaEval have an average length of 1092, for M_2 they are 1552, and for M_3 they are 2552, so the model is learning to generate longer responses, which we note may be a factor in relative performance.

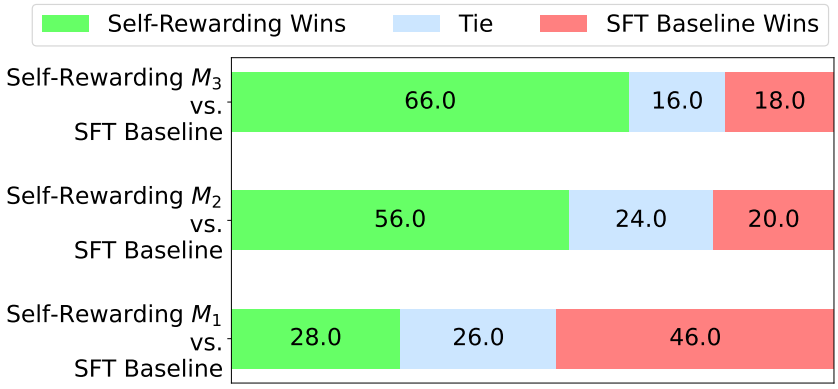

Figure 5: **Human evaluation results.** Iterations of Self-Rewarding (M_1, M_2 and M_3) provide progressively better head-to-head win rates compared to the SFT baseline, in agreement with the automatic evaluation results.

Table 2: **MT-Bench Results** (on a scale of 10). Self-Rewarding iterations yield improving scores across various categories. Math, code & reasoning performance and iteration gains are smaller than for other categories, likely due to the makeup of the Open Assistant seed data we use.

	Overall Score	Math, Code & Reasoning	Humanities, Extraction, STEM, Roleplay & Writing
SFT Baseline	6.85	3.93	8.60
M_1	6.78	3.83	8.55
M_2	7.01	4.05	8.79
M_3	7.25	4.17	9.10

Table 3: **NLP Benchmarks.** Self-Rewarding models mostly tend to maintain performance compared to the Llama 2 70B base model and the SFT Baseline, despite being fine-tuned on very different instruction-following prompts.

	ARC (↑) challenge	HellaSwag (↑)	GSM8K (↑)	MMLU (↑)	NQ (↑)
Llama 2	57.40	85.30	56.80	68.90	25.30
SFT Baseline	55.97	85.17	50.72	69.76	34.35
M_1	57.51	84.99	60.27	69.34	35.48
M_2	54.51	84.27	59.29	69.31	33.07
M_3	53.13	83.29	57.70	69.37	31.86

Human evaluation To examine whether human judgments align with automatic evaluation results, we conduct human evaluations that compare SFT baseline generations with the generations from each iteration of Self-Rewarding training, i.e., models M_1, M_2, and M_3. Specifically, we randomly select 50 instructions from the IFT test set. Each instruction corresponds to three pairs of generations (i.e., baseline vs. M_1, baseline vs. M_2, baseline vs. M_3). For each pair of generations, we assign them to three different annotators (blind evaluation performed by the authors) to make a pairwise judgment, and take a majority vote to decide which generation is better. The human evaluation results are shown in Figure 5. We find that Self-Rewarding models from later iterations show a larger advantage over the SFT baseline model, which is consistent with GPT-4's judgments, and demonstrates the effectiveness of our iterative training procedure.

MT-Bench performance further validates these results We report performance on MT-Bench in Table 2 for the SFT baseline and iterations of the Self-Rewarding model. We